

Mini Project Report

on

GreenTrace Sustainability Data Warehouse

A Data Vault 2.0 Approach for Auditable ESG Reporting and Carbon Intelligence

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Project Note

This is to certify that the mini project, entitled **GreenTrace Sustainability Data Warehouse**, is a bona fide record of the work carried out by the following students as part of the Database Management and Warehousing course requirements at IIIT Dharwad.

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Chapter 1

Introduction

Modern sustainability reporting depends on the ability to consolidate operational, environmental, and governance data into one trustworthy analytical system. In practice, however, enterprise ESG data is often fragmented across multiple formats and systems. Energy readings may arrive through meter or sensor logs, logistics activity may be distributed through CSV exports, and governance or operational indicators may be maintained through manual entry or supporting business records. This fragmentation makes audit preparation difficult, slows reporting workflows, and reduces confidence in final sustainability numbers.

The GreenTrace project addresses this problem through a sustainability-focused data warehouse built around the Data Vault 2.0 methodology. Instead of collapsing all information into a single static reporting table, the system models core business entities such as facilities, suppliers, meters, shipments, reporting periods, and carbon credits in a structured and historized warehouse. It then transforms these warehouse records into derived emission events, reporting summaries, lineage-aware metadata, and an interactive dashboard layer.

The project is motivated by modern compliance needs, especially the increasing importance of carbon accounting and ESG reporting under evolving international regulations such as the Corporate Sustainability Reporting Directive (CSRD). For such use cases, reporting is not only about producing totals. It is equally important to explain where the values came from, how they were transformed, and whether the same asset or emission has been counted more than once.

The key objectives of GreenTrace are:

- integrate fragmented ESG datasets into a consistent warehouse,
- preserve traceability and auditability across all warehouse layers,
- compute Scope 2 and Scope 3 emissions using explicit business logic,

- model carbon credits through a ledger-oriented structure,
- create a reporting-ready data mart for analytics and compliance, and
- present the results through a clear and presentation-ready interface.

The completed system demonstrates an end-to-end database and warehousing workflow. It combines schema design, ETL orchestration, analytical SQL, derived emissions logic, metadata handling, and dashboard presentation in one coherent implementation.

Chapter 2

Related Work and Design Motivation

Classical data warehousing literature provides two major design traditions. Inmon’s enterprise data warehouse approach emphasizes centralized, integrated storage of organizational data, while Kimball’s dimensional modeling approach focuses on analytics-ready schemas and reporting marts. Both approaches have shaped how modern analytical systems are designed and deployed.

ESG and carbon reporting, however, place especially strong emphasis on historization, auditability, and source traceability. Sustainability metrics often depend on data collected from multiple operational systems over time, and the reporting workflow may require a clear explanation of how raw values were transformed into final compliance metrics. For this reason, Data Vault 2.0 offers a strong conceptual fit for the GreenTrace problem.

Data Vault 2.0 separates business identifiers, relationships, and descriptive context into hubs, links, and satellites. This provides several advantages:

- stable business keys can be preserved independently of descriptive changes,
- many-to-many enterprise relationships can be modeled cleanly,
- descriptive attributes can evolve historically over time, and
- downstream analytical logic can be built without destroying source fidelity.

GreenTrace extends this warehousing foundation by including not only raw vault structures, but also business-vault logic for emissions, a carbon-credit ledger, a reporting mart, and a dashboard layer. In that sense, the project is not merely a warehouse schema design exercise. It is a complete sustainability analytics pipeline that shows how data engineering concepts can be connected to real compliance-oriented reporting.

Chapter 3

Data Acquisition and Preparation

Source Inventory

The implemented system works on an export located in `datasets/green_trace_export/green_trace_export_20260310_195112`. This export contains **25 source files**, of which **22 are CSV datasets**. The remaining files provide export metadata and supporting context.

Table 3.1: Core dataset profile used in GreenTrace.

Category	Count
Source files	25
CSV datasets	22
Facilities	25
Suppliers	12
Meters	100
Shipments	40
Carbon credit assets	10
Reporting periods	1
Energy readings	2400
Emission factors	6

The data spans multiple conceptual ESG categories:

- **Energy Consumption Data:** meter readings used for Scope 2 calculations.
- **Supply Chain and Logistics Data:** shipment activity, routes, vehicle types, and fuel usage used for Scope 3 calculations.
- **Master Data:** facilities, suppliers, reporting periods, and carbon-credit assets.

- **Reference and Metadata Data:** emission factors, reporting summaries, and lineage edges.

Preparation and Mapping

Before loading, the datasets were inspected and classified according to Data Vault 2.0 principles. Each file was mapped to its logical role in the warehouse. Hashed keys were validated, row counts were checked, and the load sequence was defined so that dependencies would be respected during ETL execution.

The logical mapping is shown below.

Table 3.2: GreenTrace warehouse layer breakdown.

Warehouse Layer	Number of Tables
Hubs	6
Links	3
Satellites	6
Reference	1
Business Vault	2
Ledger	2
Data Mart	1
Metadata	1
Total	22

The core warehouse tables include:

- hubs for facilities, suppliers, meters, shipments, carbon credits, and reporting periods,
- links for facility-meter, facility-period, and shipment-supplier-facility relationships,
- satellites for facility attributes, supplier attributes, energy readings, shipment activity, manual ESG metrics, and carbon-credit attributes,
- a reference table for emission factors,
- business-vault tables for Scope 2 and Scope 3 emission events,
- ledger tables for carbon-credit transactions and positions,
- a data mart for CSRD-style emissions reporting, and
- a metadata table for lineage edges.

The use of `load_dts`, `record_source`, and `hashdiff` columns across the vault layers is especially important because it strengthens auditability and historization.

Preparation Workflow

The data preparation stage followed a clear sequence:

1. inspect the export and verify the available files,
2. identify the role of each dataset in the Data Vault model,
3. confirm hashed keys and structural consistency,
4. standardize formats and load dependencies,
5. prepare ETL scripts for repeatable loading, and
6. verify the expected row counts prior to downstream computation.

This preparation stage ensured that the warehouse could be loaded in a controlled and auditable manner.

Chapter 4

System Architecture and Implementation

Overall Architecture

GreenTrace follows a layered architecture in which raw source data is progressively transformed into reporting outputs. The system is divided into the following major layers:

- **Raw Vault:** auditable storage of entities, relationships, and descriptive history.
- **Business Vault:** derived logic for emissions computation.
- **Carbon Ledger:** issue-retire accounting structures for carbon credits.
- **Data Mart:** reporting-oriented summaries for compliance and analysis.
- **Dashboard Layer:** visual presentation and interaction for project demonstration.

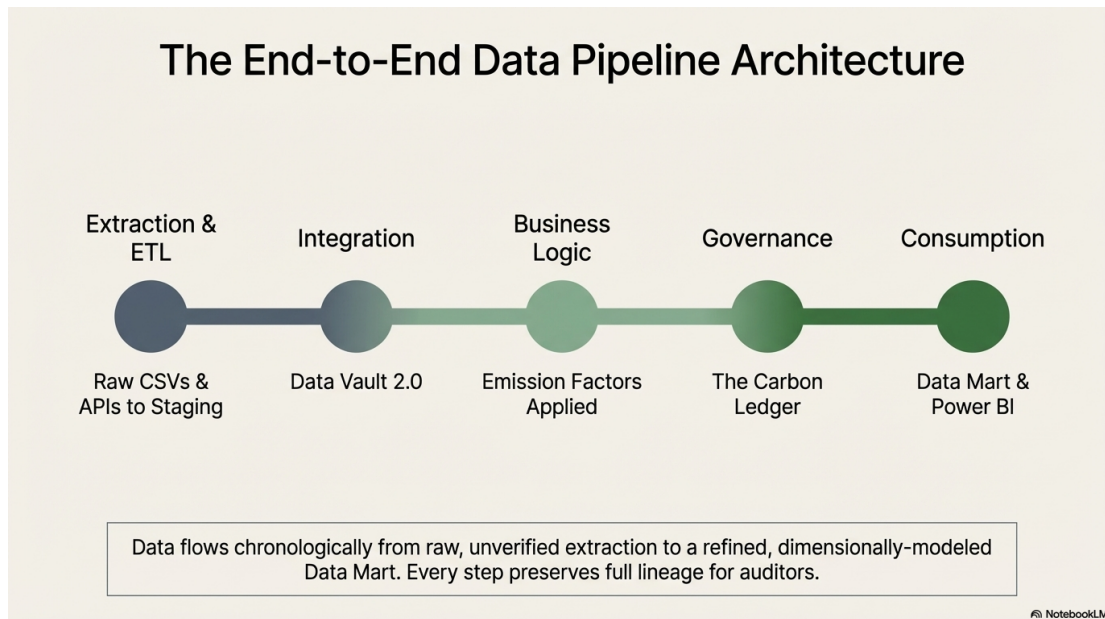


Figure 4.1: End-to-end GreenTrace pipeline from extraction and integration to governance and consumption.

Schema Design

The schema is implemented in MySQL through an idempotent SQL script. The design preserves stable business identifiers in hubs, enterprise relationships in links, and descriptive context in satellites. Primary keys, foreign keys, and indexes are used to support structural integrity and query performance.

The principal modeled entities are:

- facilities,
- suppliers,
- meters,
- shipments,
- carbon credits, and
- reporting periods.

This design supports scalability because new source records or descriptive updates can be introduced without disturbing the basic warehouse structure.

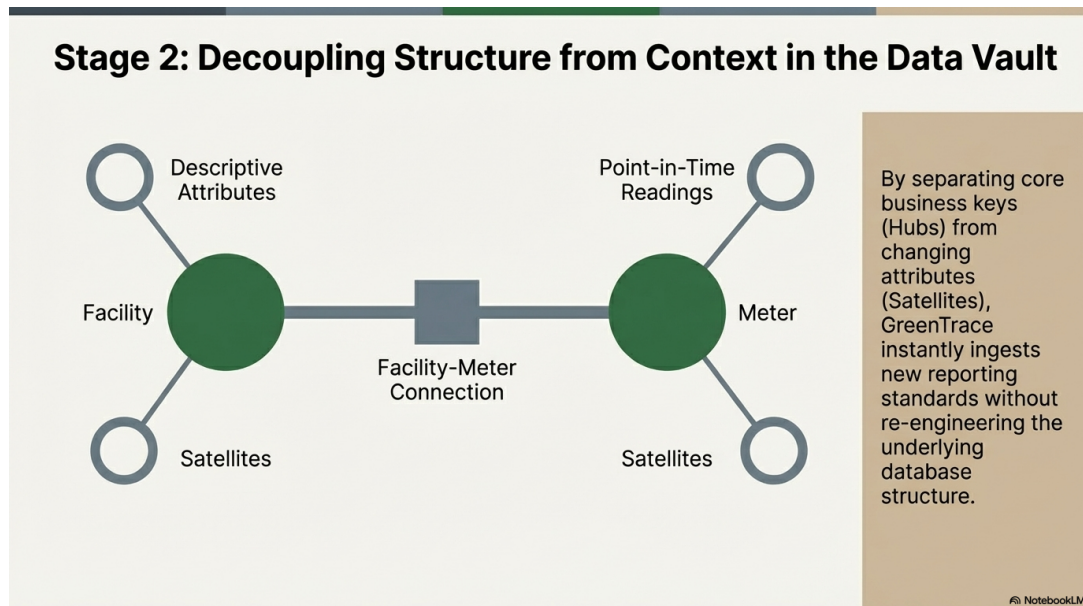


Figure 4.2: Data Vault view of how business identifiers, relationships, and descriptive context are separated for traceability.

ETL Pipeline

The ETL layer is implemented in Python using `pandas` and `SQLAlchemy`. The pipeline is dependency-aware, which means that reference tables and hubs are loaded before links, satellites, and downstream analytical tables.

The ETL process follows the classic sequence:

1. **Extract:** read CSV files from the export folder.
2. **Transform:** standardize formats, handle duplicates or missing values, and prepare consistent loading.
3. **Load:** insert records into MySQL and verify row counts.

The implemented ETL processed **5286 rows** across the 22-table warehouse. This re-runnable design is valuable both for academic demonstration and for practical warehouse maintenance.

Business Vault Computation

Once the raw vault is populated, GreenTrace computes derived emission events through the business-vault layer. The project implements two main formulas:

- Scope 2 CO2e = kWh value x emission factor
- Scope 3 CO2e = fuel used x emission factor

These computations populate:

- `bv_scope2_emission_event`
- `bv_scope3_emission_event`

Table 4.1: Business-vault output volumes.

Output Table	Rows
Scope 2 emission events	2400
Scope 3 emission events	40
Total computed events	2440

Carbon Ledger, Data Mart, and Dashboard

An important part of the project is the carbon-credit ledger. Instead of treating carbon credits as passive descriptive records, GreenTrace models them as governed assets with transaction-oriented behavior. The ledger is intended to support issuing, retiring, and auditing credits while reducing the risk of double counting.

The project also prepares a reporting layer through `dm_csrds_emissions_summary`, which aggregates emissions by facility and reporting period. This makes the warehouse suitable for downstream reporting and dashboard presentation.

The dashboard and service layer provide the final user-facing view of the project. The backend exposes routes for health checks, live analytical data, controlled editing, and limited SQL execution, while the frontend presents metrics, rankings, regional summaries, and presentation-oriented system storytelling.

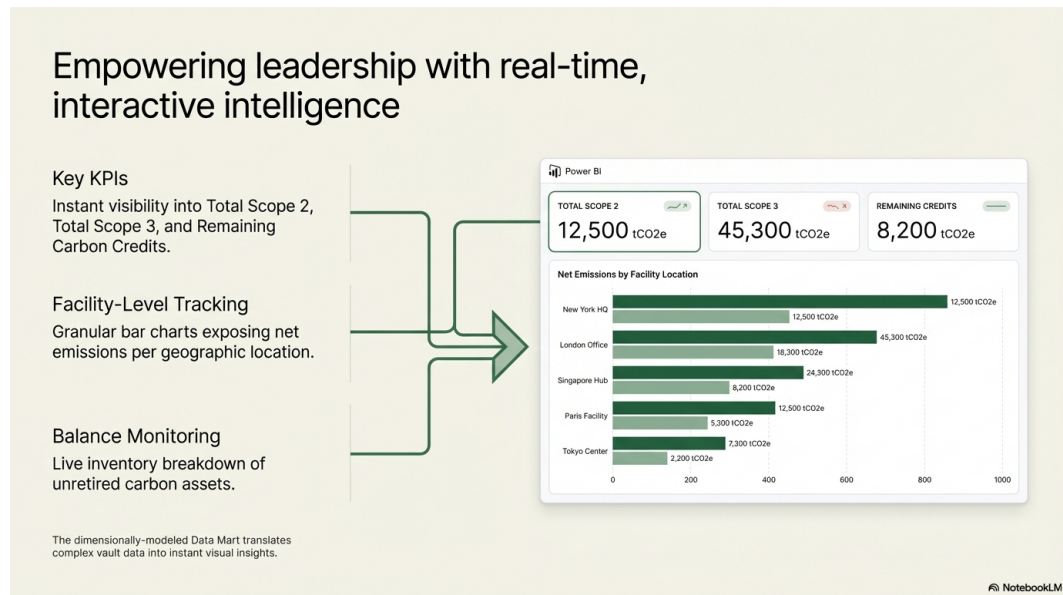


Figure 4.3: Presentation-oriented analytical interface used to communicate GreenTrace metrics and rankings.

Chapter 5

Results and Discussion

Warehouse Summary

The completed GreenTrace implementation produces a meaningful warehouse snapshot rather than a purely conceptual design. The high-level project metrics are summarized below.

Table 5.1: Overall GreenTrace warehouse snapshot.

Metric	Value
Total warehouse tables	22
Total processed rows	5286
Facilities modeled	25
Suppliers modeled	12
Meters modeled	100
Shipments modeled	40
Carbon-credit assets	10
Energy readings	2400
Reporting period	2026 Q1

Emission Results

The analytical layer reports the following emissions totals for the current snapshot.

Table 5.2: Modeled emissions in the current reporting snapshot.

Emission Metric	Value
Scope 2 emissions	40268.79 kgCO ₂ e
Scope 3 emissions	2284.99 kgCO ₂ e
Total modeled emissions (kg)	42553.78 kgCO ₂ e
Total modeled emissions (tonnes)	42.55 tCO ₂ e

The current profile shows that Scope 2 dominates the modeled footprint, contributing approximately 94.6% of the total, while Scope 3 contributes approximately 5.4%. This result is expected because the system contains a much larger volume of electricity readings than transport activity records.

Regional and Facility Insights

The reporting layer also produces operationally useful summaries.

Table 5.3: Selected analytical highlights from the GreenTrace dashboard.

Insight	Value
Top region	IN-NORTH with 9638.22 kgCO ₂ e
Peak facility	Facility 00000015 with 2006.05 kgCO ₂ e
Strongest capture interval	03:00 with 1785.31 kgCO ₂ e
Scope split	94.6% Scope 2 / 5.4% Scope 3

Table 5.4: Top five facilities by modeled total emissions.

Facility	Region	Total kgCO ₂ e
Facility 00000015	IN-NORTH	2006.05
Facility 00000005	IN-NORTH	1996.80
Facility 00000025	IN-NORTH	1933.51
Facility 00000003	IN-EAST	1894.56
Facility 00000008	IN-EAST	1891.02

Discussion

From a database management perspective, the most important strength of GreenTrace is the alignment between warehouse structure and reporting purpose. Data Vault 2.0 is used not merely as a textbook modeling idea, but as a practical way to manage fragmented ESG information while preserving traceability.

The project also demonstrates the value of connecting warehousing and presentation. The same system that stores raw energy and shipment data can produce business-vault emission events, reporting summaries, and visual insights for presentation and review. This makes the project stronger than a schema-only submission because it shows the flow from source data to interpretable results.

Another notable outcome is that GreenTrace presents sustainability reporting as a governed data problem rather than a spreadsheet problem. By modeling carbon credits, emission factors, reporting periods, and lineage-oriented transformations inside a structured warehouse, the project shows how database engineering can directly support audit-ready analytics.

Chapter 6

Conclusion

The GreenTrace Sustainability Data Warehouse demonstrates how Data Vault 2.0 can be applied to a practical ESG reporting problem. By moving from fragmented source records to an auditable and layered warehouse, the system creates a stable foundation for emissions computation, carbon-credit tracking, reporting, and dashboard presentation.

The implemented project combines core database concepts such as schema design, keys, relationships, integrity constraints, ETL sequencing, SQL-based aggregation, and warehouse layering into one coherent solution. It therefore serves both as a technical demonstration of data engineering practice and as a meaningful application in the domain of sustainability analytics.

Future work may extend the project through richer manual ESG metrics, broader emissions coverage, more detailed lineage playback, automated test coverage, and expanded reporting scenarios. Even in its present form, however, GreenTrace clearly shows how a disciplined data warehouse can turn fragmented ESG data into structured, explainable, and decision-ready carbon intelligence.

Chapter 7

Team Contributions

The project was completed collaboratively, with each member contributing to a major area of the final system.

- **Rachith Bharadwaj T N (24BDS062):** warehouse implementation, ETL orchestration, MySQL integration, backend APIs, and end-to-end system integration.
- **Ullas G M (24BDS085):** frontend design, visualization flow, dashboard presentation, and interface-level communication of the architecture.
- **M G Shree Harsha (24BDS037):** schema design, Data Vault structuring, and data-quality preparation for the warehouse layers.
- **Rahul Patil (24BDS064):** documentation support, presentation organization, and validation of submission flow and report structure.

This division of work helped the team move from schema design to an end-to-end analytical demonstration without losing structural clarity.

References

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- Ralph Kimball and Margy Ross, *The Data Warehouse Toolkit*, Wiley.
- Corporate Sustainability Reporting Directive (CSRD) oriented literature and ESG data-integration references cited in the GreenTrace project overview.